

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

### **Reproduction in plants**

- (a) define *mitosis* as cell division giving rise to genetically identical cells in which the chromosome number is maintained and state the role of mitosis in growth, repair of damaged tissues, replacement of worn out cells and asexual reproduction
- (b) define *asexual reproduction* as the process resulting in the production of genetically identical offspring from one parent and describe **one named**, commercially important application of asexual reproduction in plants
- (c) define *meiosis* as a reduction division in which the chromosome number is halved from diploid to haploid
- (d) state that gametes are the result of meiosis (reduction division)
- (e) define *sexual reproduction* as the process involving the fusion of haploid nuclei to form a diploid zygote and the production of genetically dissimilar offspring
- (f) identify and draw, using a hand lens if necessary, the sepals, petals, stamens and carpels of **one**, locally available, **named**, insect-pollinated, dicotyledonous flower, and examine the pollen grains under a light microscope
- (g) state the functions of the sepals, petals, anthers and carpels
- (h) use a hand lens to identify and describe the anthers and stigmas of **one**, locally available, **named**, wind-pollinated flower, and examine the pollen grains under a light microscope
- (i) outline the process of pollination and distinguish between self-pollination and cross-pollination
- (j) compare, using fresh specimens, an insect-pollinated and a wind-pollinated flower
- (k) describe the growth of the pollen tube and its entry into the ovule followed by fertilisation (production of endosperm and details of development are **not** required)
- (l) investigate and describe the structure of a non-endospermic seed in terms of the embryo (radicle, plumule and cotyledons) and testa, protected by the pericarp (fruit wall)
- (m) state that seed and fruit dispersal by wind and by animals provides a means of colonising new areas
- (n) describe the external features of **one**, locally available, **named** example of a wind-dispersed fruit or seed and of **one named** example of an animal-dispersed fruit or seed
- (o) investigate and state the environmental conditions that affect germination of seeds: suitable temperature, water and oxygen
- (p) describe the uses of enzymes in the germination of seeds

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**0/N18/21/Q1**

The diagram shows the parts of a plant growing above ground.



- (a) (i) State the type of reproduction that takes place using the flower shown in the diagram.

.....

[1]

- (ii) Describe the events that must take place for the flower to develop into a fruit.

.....  
.....  
.....  
.....  
.....  
.....  
.....

[5]

- (b) The plant may also reproduce without using a flower, to produce new plants from plantlets.

State the type of reproduction that produces the plantlets.

.....

[1]

**Reproduction in plants P2 questions 5090**

**Compiled by:** [Mustafa Asif](#)

(c) Suggest the advantages to the plant of reproducing by each of the following methods:

using flowers, .....

.....

.....

using plantlets, .....

.....

.....

[3]

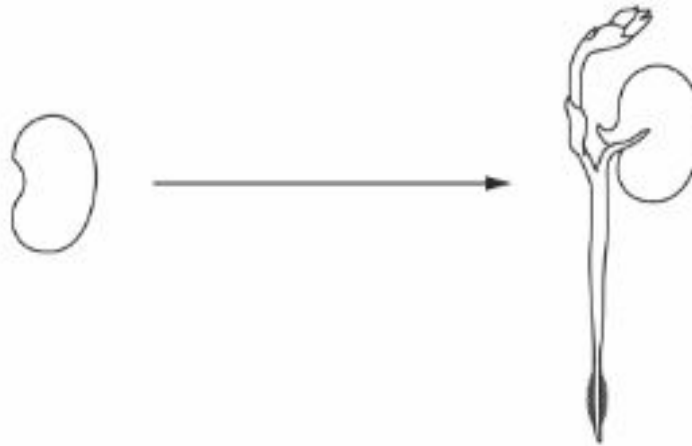
[Total: 10]

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**O/N18/21/Q2**

The diagram shows a seed (Fig. 1) and a plant developing from this seed (Fig. 2).



**Fig. 1**

**Fig. 2**

- (a) (i) Name **two** structures visible in Fig. 2 that are **not** visible in Fig. 1.

1 .....

2 .....

[2]

- (ii) Name the process represented by the arrow drawn between Fig. 1 and Fig. 2.  
Describe the use of enzymes in this process.

name of process .....

use of enzymes .....

.....  
.....  
.....

[4]

- (b) In an experiment, the growth of a plant from a seed is found to be faster as temperatures increase up to 35 °C.

Explain how the development of a plant from a seed would be affected by a temperature much higher than 35 °C.

.....  
.....  
.....

[2]

[Total: 8]

## **Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

O/N18/22/Q3

The diagram shows a pollen grain with a pollen tube growing from it.



Pollen grains from the same type of plant were placed in sucrose solutions of different concentrations for a fixed amount of time. After this time, the pollen grains and tubes were examined using a microscope. The following observations were made for each concentration of sucrose:

- the number of pollen grains that had germinated to produce a pollen tube,
- the length of each pollen tube.

The table shows the results of the investigation.

| % sucrose concentration | % of pollen grains germinated | mean pollen tube length/mm |
|-------------------------|-------------------------------|----------------------------|
| 1                       | 6                             | 0.005                      |
| 2                       | 13                            | 0.008                      |
| 4                       | 25                            | 0.015                      |
| 8                       | 56                            | 0.040                      |
| 10                      | 31                            | 0.030                      |
| 20                      | 25                            | 0.018                      |
| 40                      | 13                            | 0.006                      |

- (a) (i) A total of 12 pollen grains were placed in the 20% sucrose solution.

Use the information in the table to calculate the number of pollen grains that germinated to produce a pollen tube in the 20% sucrose solution.

..... [1]

- (ii) Suggest why the **mean** pollen tube length was calculated for each sucrose concentration.

.....  
.....[1]

**Reproduction in plants P2 questions 5090**

**Compiled by:** Mustafa Asif

- (iii) Use the information in the table to suggest the optimum (best) concentration of sucrose solution for pollen tube germination and growth.

..... %

Explain how the information in the table enabled you to reach this conclusion.

.....  
.....  
.....

[3]

- (iv) The germination of a pollen grain to form a pollen tube requires the movement of water into the pollen grain from its surroundings.

Suggest why placing a pollen grain in a solution with a higher sucrose concentration than in your answer to (a)(iii) may result in a lower percentage of germination.

.....  
.....  
.....  
.....  
.....

[3]

- (b) Describe the route taken by a growing pollen tube in a plant and explain its importance in plant reproduction.

.....  
.....  
.....  
.....  
.....  
.....  
.....

[4]

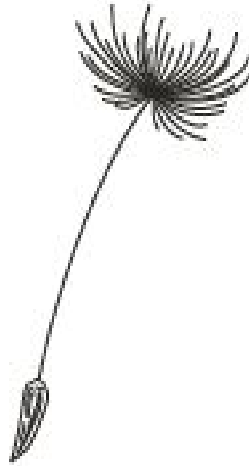
[Total: 12]

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**M/J18/21/Q3**

The diagram shows the fruit of a plant.



(a) Suggest why this fruit is most likely to be wind-dispersed.

.....

.....

.....

.....

.....[3]

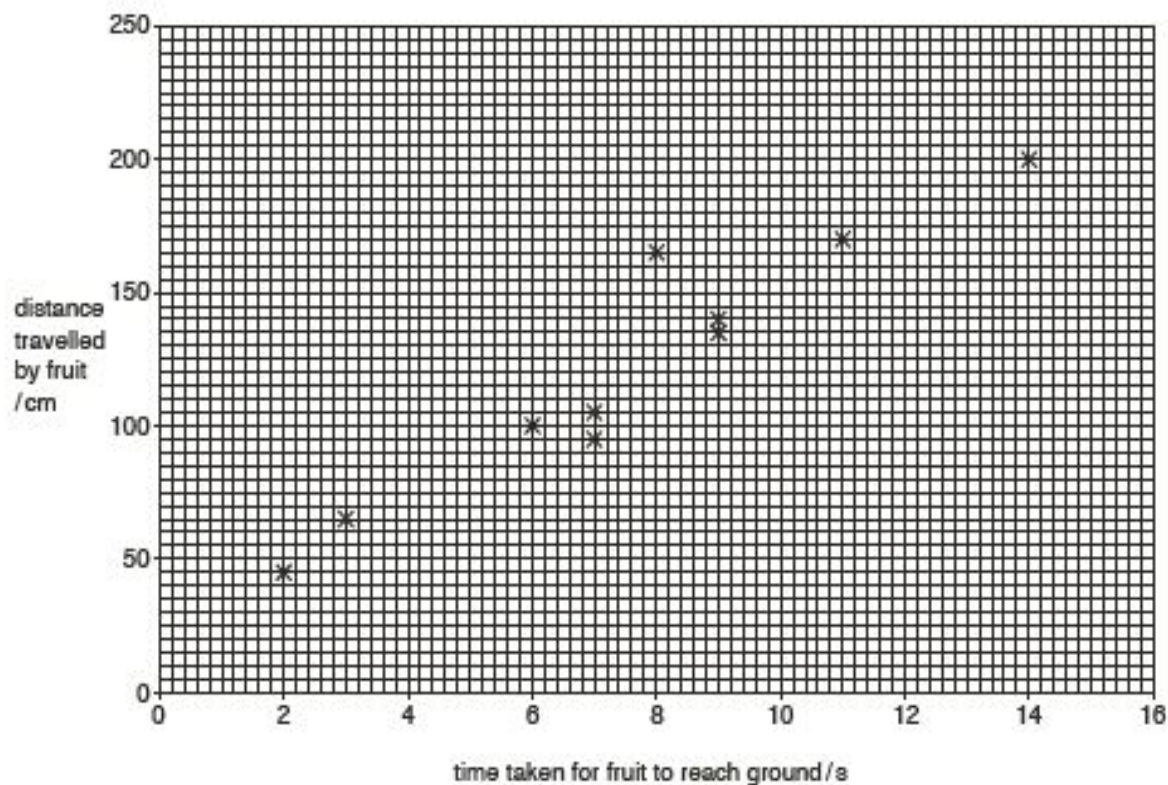


## **Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

- (b) An investigation was carried out to find the time taken for several of these fruits to reach the ground when dropped from the same height. The horizontal distance travelled by each fruit in that time was also measured.

The results of the investigation are shown in the graph.



- (i) Describe the pattern shown by the results in the graph.

.....  
.....[1]

- (ii) State the number of fruits in the investigation that took between 4 and 10 seconds to reach the ground.

.....[1]



**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

- (iii) Suggest and explain advantages to the plant of its fruits taking a longer time to reach the ground.

.....

.....

.....

.....

.....

.....[4]

- (c) When a fruit of the plant reaches the ground, the process of germination to form a new plant may take place.

State one environmental condition that affects germination and explain the importance of this condition.

environmental condition .....

explanation .....

.....[2]

[Total: 11]

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

**M/J18/22/Q1**

The diagrams show the bud of an insect-pollinated flower and a magnified transverse section through the same flower bud. The transverse section was taken at the position shown by the dotted line.

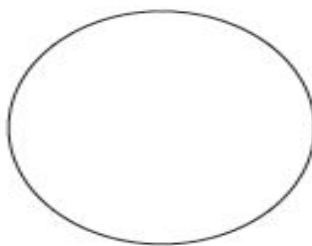


- (a) Complete the table to show the name of each of structures A to D and to state one function of each structure.

| letter | name of structure | function |
|--------|-------------------|----------|
| A      |                   |          |
| B      |                   |          |
| C      |                   |          |
| D      |                   |          |

[8]

- (b) The diagram shows an incomplete transverse section through the stem of this plant.



- (i) Complete the diagram by drawing and labelling the positions of each of the following tissues:

- xylem,
- phloem.

[2]

- (ii) State one function of xylem tissue.

.....  
.....[1]

[Total: 11]

**Compiled by:** Mustafa Asif

Fig. 5.1 shows the fruits of two plants, A and B. Both fruits are animal-dispersed.

[illegible]

The statements **E** to **K** relate to the process of reproduction.

- E** produces genetically identical offspring
- F** produces more individuals of the same species
- G** involves only one parent
- H** involves fusion of nuclei
- I** requires gametes
- J** forms a diploid zygote
- K** involves only cell division by mitosis

Table 5.1

| sexual reproduction only | asexual reproduction only | both sexual and asexual reproduction |
|--------------------------|---------------------------|--------------------------------------|
|                          | E                         |                                      |

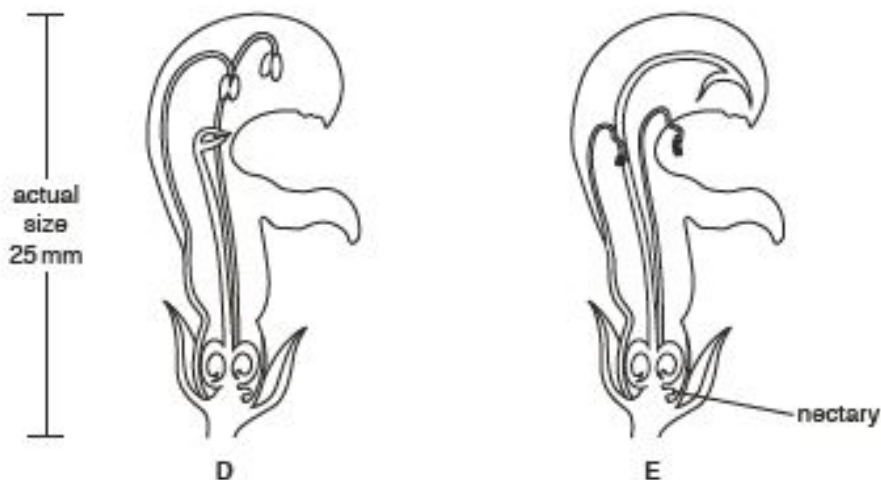
The first letter has been written in the correct box for you. Use each letter **once** only.

[6]

[Total: 6]

**M/J17/22/Q3**

Fig. 3.1 shows flowers from the same species of plant at different stages, D and E, in their development.



**Fig. 3.1**

(a) On Fig. 3.1, use lines labelled S and C to label a sepal (S) and a carpel (C). [2]

(b) The flowers are cross-pollinated by an insect. Explain why the insect must visit flower D before visiting flower E.

.....

.....

.....

..... [3]

(c) Suggest how flowers of this species are adapted to be pollinated by an insect such as a bee.

.....

.....

.....

.....

..... [4]

[Total: 9]

**Compiled by:** Mustafa Asif

(a) Fig. 7.1 shows a section through a fruit.

[illegible]

10

**FACEBOOK.COM/CAMBRIDGE O LEVEL/IGCSE(GROUP)**

**Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

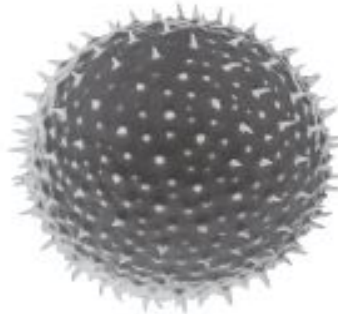
**O/N16/21/Q2**

Fig. 2.1 shows a magnified pollen grain from each of two plant species, A and B.



magnification  $\times 830$

pollen grain from plant species A



magnification  $\times 200$

pollen grain from plant species B

**Fig. 2.1**

- (a) (i) Use the information in Fig. 2.1 and your biological knowledge to describe **three** differences between the pollen grains from species A and species B.

Write your answers in Table 2.1.

**Table 2.1**

| pollen grain from species A | pollen grain from species B |
|-----------------------------|-----------------------------|
|                             |                             |
|                             |                             |
|                             |                             |

[3]

**Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

- (II) Cross-pollination takes place in both species A and species B.

Describe what is meant by the term *cross-pollination*.

.....

.....

.....

.....

.....

..... [3]

- (III) Using the information provided by Fig. 2.1, describe how cross-pollination is most likely to occur in species A and in species B.

species A

.....

.....

.....

.....

species B

.....

.....

.....

..... [3]

- (b) Fig. 2.2 is a photograph of the flowers of species A.



**Fig. 2.2**

List **two** features that would be present in the flowers of species B that are **not** present in those of species A.

- 1 .....
- 2 ..... [2]

[Total: 11]



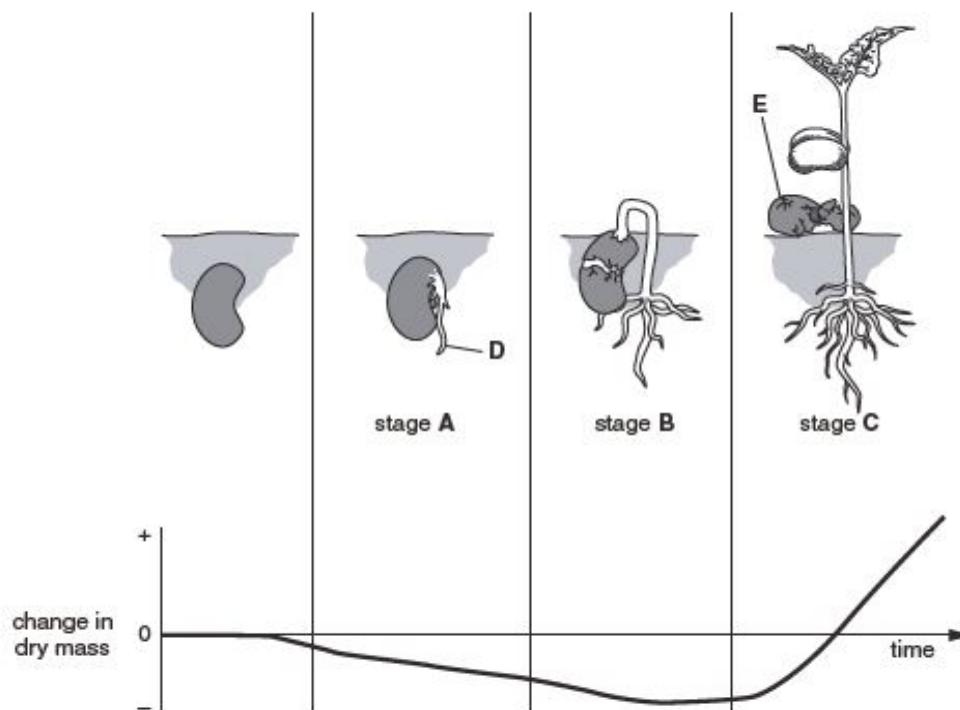
## **Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**M/J16/21/Q2**

Dry mass is the mass of all chemicals, excluding water, present in an organism.

Fig. 2.1 shows the stages of germination of a seed. Fig. 2.1 also shows the changes in dry mass of the plant during these stages of germination.



**Fig. 2.1**

(a) Name the parts labelled D and E in Fig. 2.1.

D .....

E .....

[2]

**Reproduction in plants P2 questions 5090**

**Compiled by:** Mustafa Asif

- (b) Describe and explain the changes in dry mass shown during each of the stages A, B and C in Fig. 2.1.

stage A .....

.....

.....

.....

.....

stage B .....

.....

.....

.....

.....

stage C .....

.....

.....

.....

..... [6]

- (c) Water is needed for germination of seeds.

State two other conditions needed for germination. Explain why each condition is needed.

condition .....

explanation .....

.....

.....

condition .....

explanation .....

.....

..... [4]

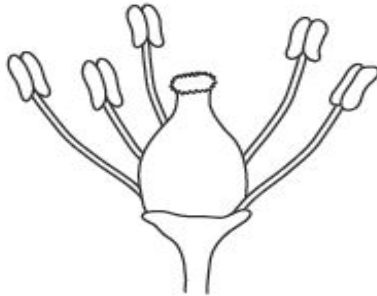
[Total: 12]

## **Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**M/J16/22/Q1**

The grape plant has many flowers clustered together on each flowering head. Fig. 1.1 shows the structure of one mature grape flower.



**Fig. 1.1**

- (a) (i) Name two parts, normally found in other flowers, that are missing from this mature flower.

1 .....

2 .....

[2]

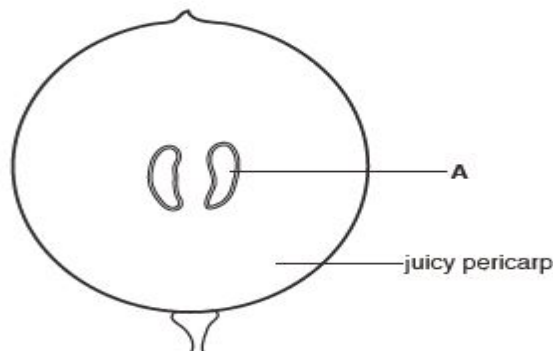
- (ii) Suggest the type of pollination found in the grape plant and give a reason for your choice.

type of pollination .....

reason .....

.....[2]

- (b) Fig. 1.2 shows a section through a fruit that develops from the same flower.



**Fig. 1.2**

- (i) Name the structure labelled A in Fig. 1.2. ....[1]

- (ii) Suggest how part A is dispersed and give reasons for your answer.

.....

.....

.....

.....

.....

.....

.....[3]

O/N15/22/Q1

Fig. 1.1 shows a carpel of a flower with two pollen grains on top.

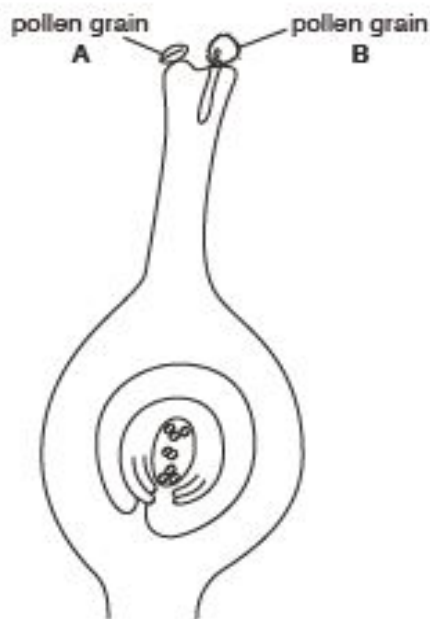


Fig. 1.1

(a) Name each of the following:

- (i) the process that has brought the pollen grains to the carpel

.....[1]

- (ii) the part of the carpel to which the pollen grains are attached.

.....[1]

(b) On Fig. 1.1, draw a line to show the route taken by the pollen tube up to the point at which male and female gametes fuse. [3]

(c) Name each of the following:

- (i) a part of the carpel where the nuclei are all **diploid**

.....[1]

- (ii) a part of the carpel that contains **haploid** nuclei.

.....[1]

(d) Suggest why pollen grain A has not developed a pollen tube.

.....

.....[1]

[Total: 8]

## Reproduction in plants P2 questions 5090

**Compiled by:** Mustafa Asif

**O/N15/22/Q9**

- (a) Outline the processes of *mitosis* and *meiosis*, including references to where they occur.

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |   |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 | 201 | 202 | 203 | 204 | 205 | 206 | 207 | 208 | 209 | 210 | 211 | 212 | 213 | 214 | 215 | 216 | 217 | 218 | 219 | 220 | 221 | 222 | 223 | 224 | 225 | 226 | 227 | 228 | 229 | 230 | 231 | 232 | 233 | 234 | 235 | 236 | 237 | 238 | 239 | 240 | 241 | 242 | 243 | 244 | 245 | 246 | 247 | 248 | 249 | 250 | 251 | 252 | 253 | 254 | 255 | 256 | 257 | 258 | 259 | 260 | 261 | 262 | 263 | 264 | 265 | 266 | 267 | 268 | 269 | 270 | 271 | 272 | 273 | 274 | 275 | 276 | 277 | 278 | 279 | 280 | 281 | 282 | 283 | 284 | 285 | 286 | 287 | 288 | 289 | 290 | 291 | 292 | 293 | 294 | 295 | 296 | 297 | 298 | 299 | 300 | 301 | 302 | 303 | 304 | 305 | 306 | 307 | 308 | 309 | 310 | 311 | 312 | 313 | 314 | 315 | 316 | 317 | 318 | 319 | 320 | 321 | 322 | 323 | 324 | 325 | 326 | 327 | 328 | 329 | 330 | 331 | 332 | 333 | 334 | 335 | 336 | 337 | 338 | 339 | 340 | 341 | 342 | 343 | 344 | 345 | 346 | 347 | 348 | 349 | 350 | 351 | 352 | 353 | 354 | 355 | 356 | 357 | 358 | 359 | 360 | 361 | 362 | 363 | 364 | 365 | 366 | 367 | 368 | 369 | 370 | 371 | 372 | 373 | 374 | 375 | 376 | 377 | 378 | 379 | 380 | 381 | 382 | 383 | 384 | 385 | 386 | 387 | 388 | 389 | 390 | 391 | 392 | 393 | 394 | 395 | 396 | 397 | 398 | 399 | 400 | 401 | 402 | 403 | 404 | 405 | 406 | 407 | 408 | 409 | 410 | 411 | 412 | 413 | 414 | 415 | 416 | 417 | 418 | 419 | 420 | 421 | 422 | 423 | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 | 432 | 433 | 434 | 435 | 436 | 437 | 438 | 439 | 440 | 441 | 442 | 443 | 444 | 445 | 446 | 447 | 448 | 449 | 450 | 451 | 452 | 453 | 454 | 455 | 456 | 457 | 458 | 459 | 460 | 461 | 462 | 463 | 464 | 465 | 466 | 467 | 468 | 469 | 470 | 471 | 472 | 473 | 474 | 475 | 476 | 477 | 478 | 479 | 480 | 481 | 482 | 483 | 484 | 485 | 486 | 487 | 488 | 489 | 490 | 491 | 492 | 493 | 494 | 495 | 496 | 497 | 498 | 499 | 500 | 501 | 502 | 503 | 504 | 505 | 506 | 507 | 508 | 509 | 510 | 511 | 512 | 513 | 514 | 515 | 516 | 517 | 518 | 519 | 520 | 521 | 5 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|

- (b) Describe the advantages of asexual reproduction in plants with reference to a named commercially important application.

.....[3]

[Total: 10]

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**M/J15/21/Q8**

- (a) Describe the external features of **one named** example of a wind-dispersed fruit or seed. State the importance to the plant of this method of dispersal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- (b) State **two** environmental conditions that affect the germination of seeds. Describe the importance of each condition.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]



**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**M/J15/22/Q6**

(a) For a named fruit or seed, describe how it is adapted for animal dispersal.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) A student planted seeds from different types of plant in the same area of soil.

(i) Suggest why some of the seeds did not germinate.

.....

.....

.....

..... [2]

(ii) Explain why several of the seedlings were unable to survive after a few weeks.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]



**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**O/N14/21/Q8**

- (a) (I) Explain how meiosis is involved in the formation of gametes.

.....

.....

.....

..... [2]

- (II) Explain how two plants, one with red flowers and one with white flowers, can produce seeds that will grow into plants with either red or white flowers in the ratio of 1:1.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

- (b) Suggest how two plants, of a different species, one with red and one with white flowers, could produce seeds that will grow into plants with pink flowers.

.....

.....

.....

.....

..... [2]

[Total: 10]

**Compiled by:** Mustafa Asif

(a) Describe the differences between an ovule, a seed and a fruit.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1

(b) Explain how fruits or seeds may be modified for dispersal by wind.

[4]

**FACEBOOK.COM/CAMBRIDGE O LEVEL/IGCSE(GROUP)**

## Reproduction in plants P2 questions 5090

**Compiled by:** Mustafa Asif

**M/J14/21/Q7**

- (a) Compare the processes of mitosis and meiosis.

[4]

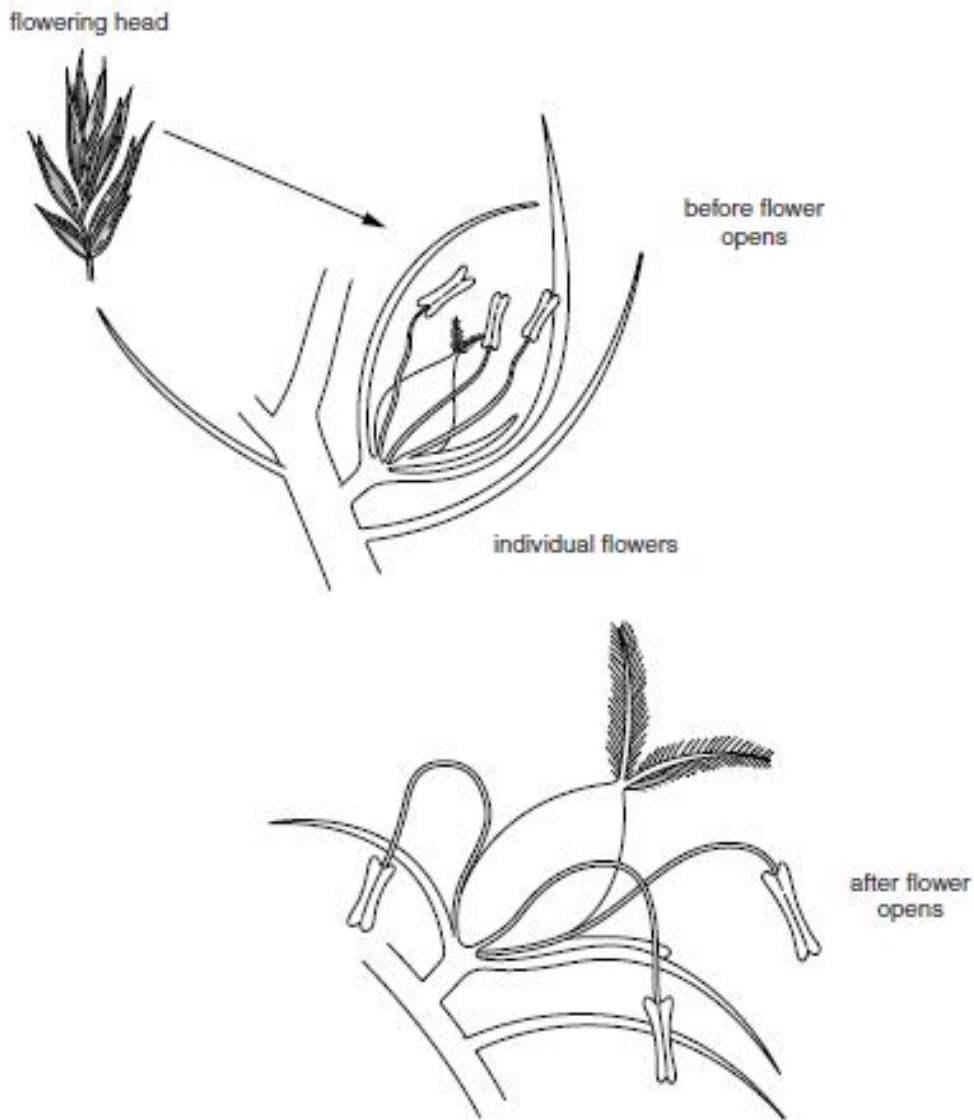
- (b) Use your knowledge of how sex is inherited to explain which parent determines the sex of a child.

[6]

[Total: 10]

**M/J14/22/Q3**

Fig. 3.1 shows a flowering head of wheat, and individual flowers before and after opening.



**Fig. 3.1**

The anthers release most of their pollen before the flower opens. The rest is released after the flower opens.

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

- (a) Name the type of pollination found in the wheat plant before the flower opens.

..... [1]

- (b) (I) Using the information provided by Fig. 3.1, describe pollination in the wheat plant after the flower opens.

.....  
.....  
.....  
.....  
..... [3]

- (II) Wheat pollen is relatively heavy and is released for only a few hours after the flowers open.

Suggest **two** disadvantages of this.

.....  
..... [2]

- (c) Scientists are working to introduce genes into wheat plants to make them resistant to attack by insect pests (greenfly) and to encourage root nodule bacteria from pea and bean plants to live in their roots.

- (I) Name the type of experimental work in which these scientists are involved.

..... [1]

- (II) Suggest how the growth of root nodule bacteria on the roots of wheat plants could reduce the amount of fertiliser required by a growing wheat crop.

.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
..... [5]

[Total:12]

**Reproduction in plants P2 questions 5090**

Compiled by: Mustafa Asif

**O/N13/21/Q8**

- (a) Explain the importance of *meiosis* in sexual reproduction.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) Fruits and seeds may be dispersed by the wind.

Describe the features of a **named** wind-dispersed fruit or seed and explain the importance of these features in the process of dispersal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

[Total: 10]



**Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

**O/N13/22/Q8(a)**

(a) Describe the part played by an ovule in the reproduction of a plant.

.....

.....

.....

.....

.....

.....

.....

.....

.....

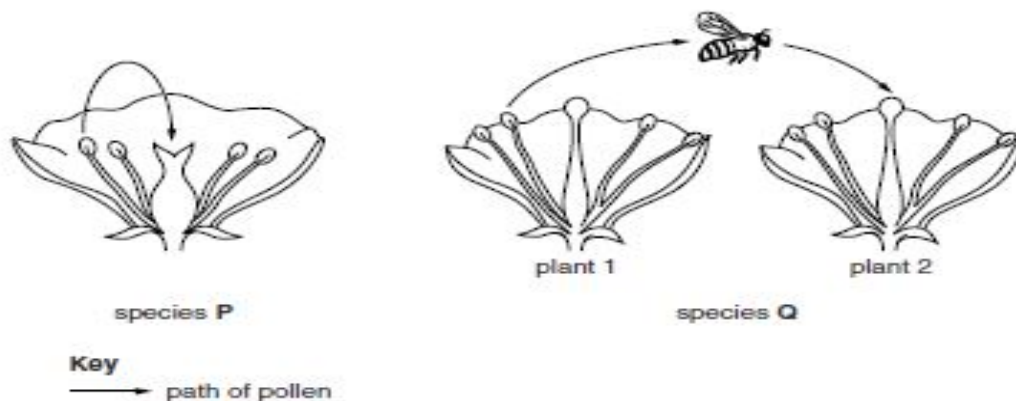
.....

[4]



M/J13/21/Q3

- (a) Fig. 3.1 shows how pollination takes place in two different species of plant, species P and species Q.



**Fig. 3.1**

- (i) Using the information in Fig. 3.1, suggest and explain how each of these species of plant is pollinated.

species P

.....

.....

species Q

.....

.....

[4]

- (ii) Suggest **two** ways in which a flower from a plant of species Q in Fig. 3.1 may be adapted to increase the likelihood of pollination taking place.

1. ....

2. ....

[2]

- (b) State why species Q shows more variation in its phenotype than species P.

.....

..... [1]

- (c) Describe the events that take place in a flower after pollination until fertilisation has taken place.

.....

.....

.....

.....

.....

[3]

[Total: 10]

## Reproduction in plants P2 questions 5090

**Compiled by:** Mustafa Asif

**M/J13/22/Q6**

- (a) Crops can be grown under controlled conditions in large buildings.

Describe and explain how such buildings can provide the conditions needed for maximum crop production.

[7]

- (b)** Suggest why seeds from crops grown in this way may produce fewer offspring and only small variations compared with crops grown naturally.

[3]

[Total: 10]

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

### **MARK SCHEME**

Mark schemes will use these abbreviations:

; separates marking points

/ alternatives

() contents of brackets are not required but should be implied

R reject

A accept (for answers correctly cued by the question, or guidance for examiners)

Ig ignore (for incorrect but irrelevant responses)

AW alternative wording (where responses vary more than usual)

AVP alternative valid point (where a greater than usual variety of responses is expected)

ORA or reverse argument

underline actual word underlined must be used by candidate

+ statements on both sides of the

+ are needed for that mark

#### **0/N18/21/Q1**

1(a)(i) sexual ; 1

1(a)(ii) pollen + stigma ;

pollen tube ;

style ;

ovary / ovule ;

female gamete / ovum ;

fertilisation or described ;

growth of pericarp / ovary wall ;

petals fall off / stamens or stigma or style withers AW ;

5 Ig self- or cross-pollination

1(b) asexual / vegetative ; 1

1(c) (using flowers) maximum 2 marks from

variation AW ;

population more likely to survive environmental change / disease ;

## **Reproduction in plants P2 questions 5090**

**Compiled by:** [Mustafa Asif](#)

more widely dispersed ;

reduction of competition ;

(using plantlets) maximum 2 marks from

quick / only one plant needed / no other organism needed ;

supply of water / nutrition from parent plant ;

higher survival rate ;

new plants in environment to which plant is well suited AW

### **O/N18/21/Q2**

2(a)(i) radicle / root ;

root hairs ;

plumule / shoot / stem ;

leaves ;

hypocotyl ;

2(a)(ii) (name of process)

germination / growth ; (use of enzymes) maximum 3 marks from

digest / breakdown / hydrolysis ; amylase ;

starch + to maltose / glucose ;

protease ;

protein

+ to amino acids ;

lipase ;

fats + to fatty acids and glycerol ;

4

2(b) development AW slower / would not germinate / would not grow ;

denature ; reactions / digestion / hydrolysis / breakdown + slow / don't occur ;

### **O/N18/22/Q3**

3(a)(i) 3 ; 1 R 3%

3(a)(ii) some may not germinate / some may not produce a pollen tube ;

accuracy / precision / reliability / validity / reduce error ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

3(a)(iii) 8 ;

highest AW + % / number ;

highest AW

+ length OR longest ;

3(a)(iv) water

+ concentration / potential ;

less / no

+ water enters OR water leaves ;

osmosis / diffusion ;

3(b) stigma ;

style ovary / ovule / ovum / embryo sac / female gamete ;

movement of AW + male gamete / male nucleus / pollen nucleus ; fertilisation / fusion of gametes ;

### **M/J18/21/Q3**

3(a) reference to structures on top of fruit ;

increases surface area ;

delays descent / wind catches beneath structures / floats **AW** ;

3(b)(i) time taken increases + distance travelled increases **OR** proportional ;

3(b)(ii) 6 ; **1**

3(b)(iii) (fruits / seeds) will travel further ;

away from plant ;

away from other fruit / seeds / scattered ;

colonise other areas ;

reduced competition / prevent overcrowding ;

competition for light / water / nutrients ;

increased chance of + new plant / reproduction / germination **AW** ;

3(c) *environmental condition* temperature / water / oxygen ;

*explanation* ;

☐ (temperature) reference to enzyme activity

☐ (water) reference to solution / softening / rupture of testa / enzyme activity / transport

☐ (oxygen) reference to respiration

### **M/J18/22/Q1**

1(a)

**letter name of structure function**

**(A)** sepal / calyx ; protect ;

**(B)** petal / corolla ; attract / landing platform ;

**(C)** stamen / anther ; **or**

filament ;

produces / contains **AW** +

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

pollen / male gamete / male nucleus ;

support anther ;

**(D)** ovary / carpel / pistil ; produces / contains **AW** +

ovum / ovule / egg / female gamete ;

**or**

forms **AW** fruit / site of fertilisation ;

1(b)(i) **1** one or more vascular bundles + each oval shaped + location correct ;

'xylem' labelled on inside + 'phloem' labelled separately on outside of at

least one oval vascular bundle ;

1(b)(ii) **1** transports / carries **AW** + water / ions / minerals ;

support ;

**O/N17/21/Q5**

5 (*fruit A*)

hooks / spikes **AW** ;

on animal / fur ;

falls off / rubbed off / leaves animal **AW** ;

(*fruit B*)

edible / eaten / food ;

colour / taste / juicy / sweet ;

thrown away / discarded / faeces / egested **AW** ;

(*for A or B*)

new location / away from parent plant ;

**M/J17/21/Q5**

5 (*sexual only*)

**H** ;

**I** ;

**J** ;

**3 R** letters in more than one box

**lg** letters in incorrect box

(*asexual only*)

(**E**)

**G** ;

**K** ;

**2**

(*both*)

**F** ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

### **M/J17/22/Q3**

(a) line ending either on or in sepal + labelled either 'S' or 'sepal' ;

line ending either on or in carpel + labelled either 'C' or 'carpel' ;

3(b) (flower **D**)

anther(s) / stamen(s) ;

pollen + mature / produced / present / released / collected ;

stigma + closed / immature **AW** / does not receive pollen ;

(flower **E**)

stigma + open / mature / receives pollen ;

anthers + withered **AW OR** reference to no pollen produced / present / released / to be collected ;

**3**

3(c) large + petal(s) ;

reference to insect / bee landing **AW** on petal ;

colour(ful) + petal(s) ;

reference to scent / smell ;

reference to nectar(ies) / (nectar) guides ;

pollen + stick(y) / hooked (to attach to insect / bee) ;

insect / bee makes contact **AW** with anther / stigma + while collecting nectar ;

### **M/J17/22/Q7**

7(a) (seed)

**1** seed + results from fertilisation **AW** / sexual reproduction ;

**2** (fusion / fertilisation of) gametes **OR** pollen nucleus + egg / ovum / ovule ;

**3** reference to (gametes often) from different plants / flowers ;

**4** (gametes produced by) meiosis / reduction division ;

**5** combination **AW** of (different) genes / alleles ;

(fruit)

**6** fruit (wall) + results from mitosis ;

**7** reference to (genetically) identical / clones ;

**4 A** maximum of **three** of marking points **1** to **5**

(i.e. for 'seed')

7(b) (wind) (animal)

**1** small **OR** large ;

**2** light / buoyant **OR** heavy ;

**3** wings / parachutes **OR** hooks / hairs ;

**4** large surface area **OR** attachment (to animal) ;

**5** large number **OR** small number ;

**6** not coloured / dull **OR** colour / named colour / bright ;

**7** not sweet/juicy/flesh(y)/scented **OR** sweet / juicy / flesh(y) / scented

;

**8** reference to

food / eaten / digested ;

**9** seed + enzyme resistant

/ cannot be digested / is

egested / is lost in faeces / is

thrown away ;

### **O/N16/21/Q2**

2(a)(i) **A** small + **B** large / **A** smaller / **B** larger ;

**A** smooth + **B** spikes **AW** ;



## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

**A** wind dispersal / **B** animal dispersal ;

**A** light / **B** heavy ;

2(a)(ii) transfer / movement + of pollen ;

from anther ;

to stigma ;

of another plant (of same species) ;

**A** stamen / pollen sac

**R** another species

2(a)(iii) (plant species **A**) wind ;

reference to light / small / carried (by wind) ;

(plant species **B**) insect ;

reference to attachment (to insect) / carried (by insect)

moving between flowers ;

2(b) large (petals) ;

(brightly) coloured / patterned (petals) ;

anthers enclosed in flower / not hanging out / erect ;

scent ;

nectar ;

### **M/J16/21/Q2**

2 (a) radicle / root ;

testa / seed coat ;

[2]

(b) stage A:

decreases ;

stage B:

decreases ;

explanation for A / B:

food store used ;

reference to respiration ;

reference to enzyme / named

enzyme action ;

stage C:

increases ;

explanation for C:

reference to photosynthesis ;

reference to production of

named food compound ;

reference to more cells

made / tissue growth ;

maximum 3 marks for A / B and

maximum 3 marks for C

[max 6]

(c) oxygen ;

reference to respiration ;

energy + for growth ;

suitable / correct temperature ;

reference to enzyme ;

breaks down food store AW ;

### **M/J16/22/Q1**

1 (a) (i) sepals / calyx ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

petals / corolla ;

nectaries ;

[max 2]

(ii) type of pollination: self / wind ;

reason for wind: exposed / large +

stamens / anthers ;

reason for self: position of anthers

relative to stigma / carpel ;

reason for either : no petals / nectaries

to attract insects ;

[1]

[max 1]

(b) (i) seed / cotyledon ; [1]

(ii) bird / animal / herbivore ;

sweet / sugary / coloured

(skin) / juicy / succulent / taste / smell ;

eaten / consumed / food ;

spits out or drops / undigested / passes

out with faeces ;

at a distance from parent

plant / elsewhere AW ;

R reference to excretion

[max 3]

(ii) sugar (or named) / correct formula ;

fermenter/ fermentation / anaerobic

respiration (or description of) ;

reference to suitable

temperature / warmth ;

reference to enzymes ;

(c) (i) yeast / fungus ;

### **O/N15/22/Q1**

1 (a) (i) pollination ; [1] I insect / wind / cross / self

(ii) stigma ; [1]

(b) line (continuing) down style ;

entering micropyle ;

touching / entering embryo sac ;

[3]

(c) (i) stigma / style / ovary wall ; [1]

(ii) ovary / ovule / embryo sac / ovum / ova / egg (cell) / gamete ; [1]

(d) from a different species / type / kind of plant ;

ref. chemicals on stigma unsuitable ;

couldn't germinate / not viable / infertile AW ;

not there long enough / grain immature / wrong enzymes ;

### **O/N15/22/Q9**

9 (a)

both cell division ;

(mitosis)

in body cells or named ;

for growth / repair / replacement ;

chromosome number retained AW ;

ref. asexual reproduction / single parent ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

produces copy / identical / similar ;  
produces two cells ;  
(meiosis)  
in ovaries\* / ovule ;  
testes\* / anthers ;  
ref. gamete / sex cell production ;  
chromosome number halved / cells become haploid ;  
sexual reproduction ;  
genetically different ;  
produces four cells ;

[max 7]

max 4 marks for mitosis or meiosis

A in zygote

I growth / repair within a cell

\* OR A gonads for one mark

I fertilisation

(b) name of correct organism / crop / technique ;

always an exact / similar copy ;

known characteristics / flavour AW ;

more certain outcome / can be conducted in controlled conditions;

not reliant on pollination (agent) ;

only one parent needed ;

quick(er) ;

less expensive / greater profit / higher yield / more offspring ; A cutting, micro-propagation, tubers, bulbs, layering, runners

**M/J15/21/Q8**

8 (a) example named ;

large surface area ;

related adaptation, e.g. wing, air

bladder, etc. ;

allows plant to colonise new areas ;

reduces competition ;

(b) water ;

softening testa / seed coat ;

activation of enzymes ;

solvent ;

oxygen ;

ref. respiration\* ;

energy (for germination / growth)\* ;

suitable temperature ;

ref. enzyme / named enzyme action ;

break down food store ;

ref. respiration\* ;

energy (for germination / growth)\* ;

**M/J15/22/Q6**

6 (a) 1. name of fruit or seed ;

2. adaptation that brings animal and plant into contact ;

3. how fruit or seed is carried away ;

4. how fruit or seed is discarded by

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

animal ;

5. ref to distance from parent

plant / another location ;

1. must fit the description if one is given

2. e.g. taste, colour, hooks, etc.

3. e.g. internally, in intestine, eaten, attached to fur

4. e.g. defaecation, scratched off / falls off / spat out

[max 4]

(b) (i) 1. seeds not viable / dormant / AW ;

2. temperature not suitable ;

3. (amount of) water ;

4. some of the seeds may require light ;

5. pH ;

[max 2]

(ii) 1. ref. to competition ;

2. accept two from (for) light / nutrients / water ;

4. ref. unsuitable temperature / pH ;

5. diseases ;

6. eaten (by herbivores / insects / pests) / damage by animals ;

7. toxins in soil ; 1. A overcrowding 2. Ig refs to too much Ig CO<sub>2</sub> and O<sub>2</sub>

### **O/N14/21/Q8**

8 (a) (i) in testes / anthers / ovaries ;

cell division ;

halving of chromosome numbers / haploid ;

so that diploid number is restored on fertilisation ;

[max. 2]

(ii) one (either) colour is controlled by a dominant allele;

one by a recessive allele ;

one parent heterozygous – (or described, e.g. Rr) ;

one is homozygous recessive (or described) ;

correct ref. to gametes ;

gametes correctly identified for both parents ;

how gametes pair to produce offspring in 1:1 ratio ;

[max. 6]

(all points acceptable on an annotated genetic diagram)

(b) mutation ;

ref to a named mutagen ;

possible co-dominance ;

ref. to heterozygous plants having pink flowers ;

ref. availability of certain ions (as in Hydrangea) ;

### **O/N14/22/Q6**

(a) ovule:

in ovary ;

contains female gamete ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

ref. to haploid ;  
seed:  
ovule after fertilisation ;  
ref. diploid (or with ref. fruit) ;  
(grows) larger than ovule ;  
stores food / ref. cotyledons ;  
contains (rest of) embryo / radicle + plumule ;  
(surrounded by) testa ;  
ref. dispersal ;  
fruit:  
seed(s) + ovary (wall) / pericarp ;  
ref. dispersal ;

(b) dry ;  
light ;  
(may be) winged / hairy / feathery / helicopter / parachute ;  
large surface area ;  
to allow wind to detach it from parent plant ;  
to delay its descent ;  
(allow it to be carried) long distance AW / away from parent plant ;

### **M/J14/21/Q6**

7 (a) Mitosis Meiosis  
2 daughter cells 4 daughter cells  
haploid diploid  
chromosome number  
maintained  
chromosome number  
halved  
occurs in all organs / body  
cells  
occurs in gamete producing  
organs / named  
produces body cells / used in  
growth  
produces gametes / named  
ref. asexual reproduction ref. sexual reproduction  
no genetic variation in  
offspring AW  
genetic variation in offspring  
AW

[4]

All points must be comparative to gain credit.

(b) Father ;

Max 5 from the following:

father is XY / contains Y chromosomes ;  
mother is XX / does not contain Y chromosomes ;  
father produces sperm with either X or Y (chromosomes) ;  
mother produces eggs only with X (chromosomes) ;  
sex depends on which sperm fertilises the egg ;  
1:1 male : female in offspring / AW ;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

### **M/J14/22/Q3**

3 (a) self (–pollination) ; [1]

(b) (i)

(carried by) wind ;

pollen to stigma ;

of another (wheat) plant / flower ;

correct ref. to cross-pollination (now being possible) ;

[3] Ig ref. to animals

(ii) wind can't carry / can't be carried far / reduced dispersal ;

too much dependence on self-pollination / lack of (genetic) variation AW ;

wind may not be blowing (over short time period) ;

reduces chances of pollination / fertilisation ;

(c) (i) genetic engineering / genetic modification ;

[1] Ig gene transfer / biotechnology

(ii) (bacteria) fix / convert / change / turn ;

atmospheric / soil nitrogen ;

(to) ammonium ;

(to) nitrates ;

(to make) amino acids / proteins ;

(nitrates) absorbed / (amino acids or proteins) used by plants

### **O/N13/21/Q8**

8 (a) cell division;

ref. to chromosomes;

diploid to haploid;

gamete formation/ named gametes;

all gametes genetically different;

fertilisation restores chromosome number / AW;

leads to variation in population;

[max 4]

A halving of chromosome number

(b) name of fruit / seed (not linked to following marks);

structural feature of fruit/seed that is a wind-dispersal adaptation;

ref. large surface area / buoyancy;

\*wind removes fruit / seed from parent plant;

\*seed relatively light in mass;

\*to delay descent of seed AW;

spread over large area / prevent overcrowding / competition ;

[max 6]

Accept \* points for descriptions

of pollen for max 3

e.g. floats for long time /

increases air resistance

### **O/N13/22/Q8(a)**

8 (a) contains female gamete / female nucleus / egg cell;

in ovary;

## **Reproduction in plants P2 questions 5090**

Compiled by: [Mustafa Asif](#)

fertilized / fuse;  
(by the) male gamete / male nucleus;  
(to form) zygote / embryo;  
surrounded by integuments/testa;  
seed + germinates (to form new plant);

### **M/J13/21/Q3**

3 (a) (i) (species P)  
self / insect / wind ;  
(pollen) from anther to stigma\*;  
of same flower;  
(species Q)  
Cross;  
(by) insect (pollination);  
(pollen) from anther to stigma\*;  
of different plant;  
[max 4]

\* accept once only in either P or Q

Accept plausibly named insect eg  
bee, wasp

(ii) brightly coloured petals;  
large petals;  
scent;  
nectar;  
sticky pollen or stigma;  
[max 2]

(b) Cross pollination / has 2 parents / ref genes / alleles; [1] Accept "exchange of traits"

(c) pollen tube;  
grows / digests + down style;  
to ovule / embryo sac / micropyle;  
male + female gametes /egg / nuclei fuse;

### **M/J13/22/Q6**

(a) 1. addition / availability of carbon dioxide;  
2. controlled / optimum AW temperature (or any reasonable stated temperature);  
3. ref. light (intensity);  
4. ref. blinds during day / artificial lights (at night time);  
5. keep well supplied with water / ref. irrigation / humidity control;  
6. addition of fertiliser / any named ion / pH control / hydroponic techniques ;  
7. nitrate + protein manufacture / magnesium + chlorophyll production;  
8. photosynthesis (A anywhere relevant);  
9. growth;  
10. maximum rate / day and night / 24 hrs per day ;  
11. pest control;  
12. protection from (adverse) climatic factors or any named AW; Max 7  
Ignore refs. to O<sub>2</sub>  
A any named ion + function  
R chloroplasts



## **Reproduction in plants P2 questions 5090**

**Compiled by:** [Mustafa Asif](#)

Must be ref. P/S or growth

(b) isolation from other varieties of the species;

limited genetic variation;

can pollinate only with plants in the building / cannot cross pollinate  
with plants outside;

exclusion from agents of pollination / wind / insects;

seeds less viable;